Two-zone tapered-disc model: physical setup and optimisation

Scope. This document describes the nine configured experiments in `/fred/oz061/kandrych/smac/1_tapered_2zones_dubrulle/lower_precision/seed1`, using their configuration files, launch scripts, and the inspected modelling code. All nine experiments use the same physical model and the same 18 free parameters, with different combinations of observational constraints. The values below describe the current configured experiments rather than best-fitting models.

1 Modelling workflow

The framework connects a parameterised physical model to a selected set of observations (Fig. 1). MCFOST predicts the spectral energy distribution and images. Observable-specific processing converts these predictions into quantities that can be compared with the data. The resulting discrepancies are combined into a score that guides the next parameter proposal from SMAC. The same workflow supports different choices of physical parameterisation and observational constraints.

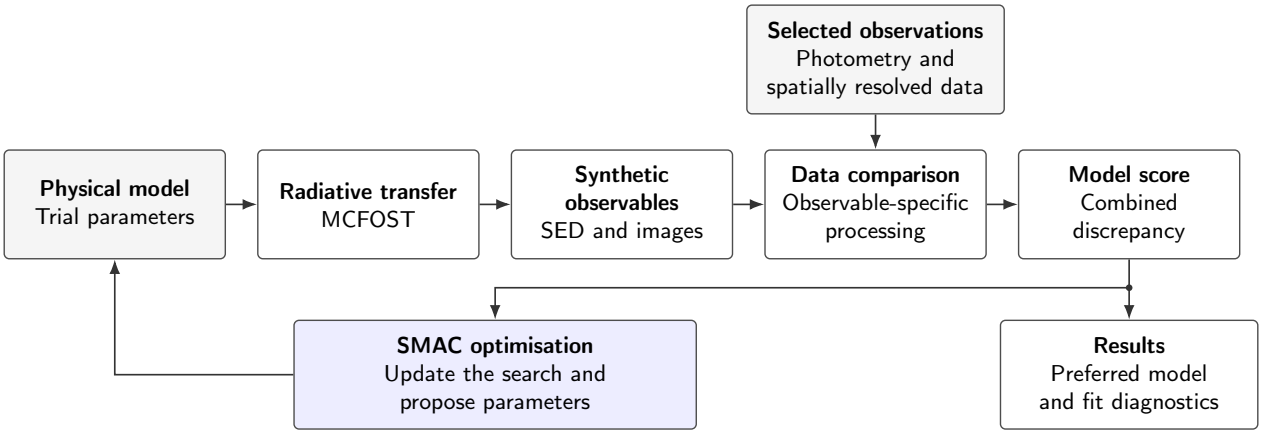


Figure 1: Overview of the modelling and optimisation framework. A trial physical model is evaluated with MCFOST and compared with the selected observations after observable-specific processing. The combined discrepancy guides the SMAC parameter search. The final outputs include the preferred model and diagnostics for the individual observables.

2 Physical model setup

We model the circumbinary disc using MCFOST, adopting an axisymmetric density structure with two contiguous radial zones. The inner zone follows a power-law surface-density distribution, while the outer zone combines a power law with an exponential taper. Both zones share the same prescribed scale-height law and dust population. Their masses are determined jointly from the total dust mass and the requirement that the surface density remains continuous at the transition radius.

2.1 Central source and system properties

The central radiation source is represented by a single star at the origin. Its properties and the system distance remain fixed throughout the optimisation (Table 1).

Table 1: Fixed system and stellar properties, with the derived stellar luminosity.

Group	Quantity	Adopted value	Treatment
System	Distance, d	1220 pc	Fixed
Central source	Effective temperature, $T_{\rm eff}$	7250 K	Fixed
Central source	Stellar radius, $R_\star$	$65\,R_\odot$	Fixed
Central source	Mass supplied to MCFOST, $M_\star$	$2\,M_\odot$	Fixed
Central source	Position, (x, y, z)	$(0, 0, 0)$ au	Fixed
Central source	Irradiating sources in MCFOST	1	Fixed
Stellar spectrum	Atmosphere file	`Kurucz7250-1.0.fits.gz`	Explicit input spectrum
Stellar spectrum	Additional UV fraction, $f_{\rm UV}$	0	Fixed; UV slope of 2.2 is inactive
Derived quantity	Stellar luminosity, $L_\star$	$\simeq 1.05 \times 10^4\,L_\odot$	Implied by $R_\star$ and $T_{\rm eff}$

The stellar properties are supplied by `simulation.para`; the distance is overridden by `configuration_tapered.yaml`. Source locations are listed in Appendix A.

An additional unresolved component is included when calculating the SED and interferometric observables. It has a blackbody temperature of 4000 K and contributes 3.9% of the combined intrinsic model flux at $1.65\,\mu$m:

$$\frac{F_{\rm secondary}}{F_{\rm MCFOST} + F_{\rm secondary}} = 0.039. \tag{1}$$

For interferometry, this component is represented by a point source at the image centre. It is added after radiative transfer and therefore does not contribute to the calculated disc heating.

2.2 Radial density structure

The radial structure is described by $R_{\rm in}$, $R_{\rm mid}$ and $R_{\rm out}$, the inner, transition and outer radii, respectively. Let s_1 and s_2 denote the surface-density exponents used by the code. The prescribed profiles

are

$$\Sigma(r) = \begin{cases} \Sigma_{\rm mid} \left(\dfrac{r}{R_{\rm mid}} \right)^{s_1}, & R_{\rm in} \le r \le R_{\rm mid}, \\ \Sigma_{\rm mid} \left(\dfrac{r}{R_{\rm mid}} \right)^{s_2} \exp\left[- \left(\dfrac{r}{R_c} \right)^{2+s_2} + \left(\dfrac{R_{\rm mid}}{R_c} \right)^{2+s_2} \right], & R_{\rm mid} < r \le R_{\rm out}. \end{cases} \tag{2}$$

Here, R_c sets the characteristic taper radius. The exponential normalisation ensures continuity at $R_{\rm mid}$. The adopted parameter ranges allow the inner surface density to remain constant or increase outwards, while the outer profile decreases outwards.

The launch option `-tapered-edge-p1-eq-p2` ties the two exponent fields *within the outer zone*. Consequently, the taper power is $2 + s_2$; it is not an additional free parameter. The inner and outer slopes, s_1 and s_2, remain independent.

The common normalisation satisfies

$$M_{\rm dust} = 2\pi \int_{R_{\rm in}}^{R_{\rm out}} \Sigma(r)\, r \, {\rm d}r. \tag{3}$$

The code derives the individual zone masses from their respective integrals. The transition introduces a change in slope while preserving surface-density continuity.

2.3 Vertical structure, settling and dust

Both zones use the prescribed scale-height law

$$H(r) = H_0 \left(\frac{r}{R_0} \right)^{\beta}, \qquad R_0 = 7.2 \text{ au}. \tag{4}$$

The reference radius stays fixed even when $R_{\rm in}$ varies. Thus, H_0 always denotes the scale height at 7.2 au; the height at the inner edge is $H(R_{\rm in})$.

Dust settling follows the Dubrulle prescription, with its strength controlled by α. This prescription retains Gaussian vertical profiles with grain-dependent scale heights. Viscous heating is also enabled, so α participates in both settling and heating. The vertical structure is prescribed, with the hydrostatic-equilibrium iteration disabled. These parameter interpretations follow the MCFOST disc-physics documentation.

The dust population comprises a mixture of amorphous silicate and iron, with a power-law grain-size distribution between $a_{\rm min}$ and $a_{\rm max}$. The complete population is shared between the two zones. Grain optical properties use the distribution of hollow spheres (DHS) prescription and effective-medium mixing. Material porosity and the maximum DHS void fraction, $V_{\rm max}$, are separate parameters; see the MCFOST dust-property documentation.

Table 2: Fixed physical assumptions and quantities derived from the free parameters.

Group	Assumption or dependent quantity	Adopted setup
Radial structure	Number and type of zones	Two: inner power-law disc; outer tapered disc
Radial structure	Zone boundary	$R_{\rm out,1} = R_{\rm in,2} = R_{\rm mid}$
Radial structure	Zone masses	Derived from $M_{\rm dust}$ and surface-density continuity
Radial structure	Additional edge smoothing	`edge = 0` in both zones

Group	Assumption or dependent quantity	Adopted setup
Outer taper	Taper power	$2 + s_2$, tied to the outer density slope
Vertical structure	Reference radius	$R_{0,1} = R_{0,2} = 7.2$ au
Vertical structure	Scale height and flaring	$H_{0,2} = H_{0,1}$, $\beta_2 = \beta_1$
Vertical structure	Settling prescription	Dubrulle; `dust_settling = 2`
Disc physics	Gas-to-dust mass ratio	100 in both zones
Disc physics	Dust radial migration	Enabled
Disc physics	Viscous heating	Enabled
Disc physics	Dust sublimation calculation	Disabled
Disc physics	Hydrostatic-equilibrium iteration	Disabled
Disc physics	Additional puffed-up-rim option	Disabled
Dust population	Species per zone	One composite population
Dust composition	Silicate optical constants	`sil_amorph_DIANA.lnk`
Dust composition	Iron optical constants	`Fe_palik.lnk`
Dust composition	Material fractions	$f_{\mathrm{sil}} = 1 - f_{\mathrm{Fe}}$
Dust optics	Grain shape and mixing	DHS; effective-medium mixing
Dust heating	Heating method	Radiative equilibrium and LTE
Zone coupling	Composition and grain properties	Entire zone-1 population copied to zone 2

The forsterite entry in the unmodified template is replaced by iron during parameter-file generation.

2.4 Numerical radiative-transfer settings

The radiative-transfer calculations use the fixed settings in Table 3. These control numerical sampling and resolution rather than adding physical fit parameters.

Table 3: Fixed numerical settings for the radiative-transfer calculations.

Numerical setting	Value
Photon packets for temperature calculation	1.28×10^5
Photon packets for SED calculation	1.28×10^4
Photon packets for images	1.28×10^6
Wavelength grid	50 points spanning 0.1–3000 μm
Spatial grid	Cylindrical; $n_r = 200$, $n_z = 200$, $n_\phi = 1$
Inner radial refinement setting	$n_{\mathrm{rad,in}} = 30$

Numerical setting	Value
Grain-size sampling	100 logarithmically distributed sizes
Standard image grid	1000×1000 pixels; map size 800 au
Viewing directions per model	One inclination and one azimuth
Polarisation	Stokes calculations enabled
ALMA image geometry	Set separately from the observation metadata

The `4.1` header identifies the template's parameter-file version; the running MCFOST binary version is not established by that header alone.

3 Optimisation approach

We explore a common parameter space across nine experiments to assess the constraints provided by different combinations of observations. The 18 free physical parameters describe radial structure, viewing geometry, vertical structure and settling, and dust properties. Stellar properties, the distance, and the physical prescriptions listed above remain fixed.

3.1 Parameter selection and search ranges

Table 4 lists the grouped search space. The defaults are configuration reference values, not fitted estimates or necessarily the first evaluated model. The search contains 12 continuous, four integer, and two categorical dimensions. Logarithmic sampling applies to the dust mass and grain-size limits.

Table 4: Free physical parameters shared by all nine current seed1 experiments: allowed ranges, search representations and YAML defaults.

Group	Parameter (symbol)	Allowed range or values	Representation	Default
Radial structure	Total dust mass ($M_{\rm dust}$)	10^{-4}–$10^{-1}\,M_\odot$	Continuous, logarithmic	10^{-2}
Radial structure	Inner radius ($R_{\rm in}$)	5–10 au	Continuous, linear	7.2
Radial structure	Transition radius ($R_{\rm mid}$)	10–40 au	Continuous, linear	18
Radial structure	Taper radius (R_c)	150–350 au	Integer	200
Radial structure	Outer radius ($R_{\rm out}$)	300–600 au	Integer	400
Radial structure	Inner surface-density exponent (s_1)	0–4	Continuous, linear	1.5
Radial structure	Outer surface-density exponent (s_2)	-1.99 to -0.1	Continuous, linear	-1.0
Viewing geometry	Inclination (i)	0°–50°	Integer degrees	19
Viewing geometry	Disc position angle (PA)	90°–180°	Integer degrees	150
Vertical structure	Scale height at 7.2 au (H_0)	0.5–3 au	Continuous, linear	1.4
Vertical structure	Flaring exponent (β)	0.7–2.0	Continuous, linear	1.2

Group	Parameter (symbol)	Allowed range or values	Representation	Default
Settling and heating	Viscosity parameter (α)	$\{10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$	Categorical	10^{-2}
Grain sizes	Minimum grain radius ($a_{\min}$)	0.01–1 μm	Continuous, logarithmic	0.1
Grain sizes	Maximum grain radius ($a_{\max}$)	100–10 000 μm	Continuous, logarithmic	1000
Grain sizes	Size-distribution exponent (q)	2.0–4.5	Continuous, linear	2.75
Grain structure	Maximum DHS void fraction ($V_{\max}$)	0.6–0.9	Continuous, linear	0.7
Grain structure	Material porosity ($\mathcal{P}$)	0–0.8	Continuous, linear	0.5
Composition	Iron volume fraction (f_{Fe})	$\{0, 0.05, 0.10, 0.15\}$	Categorical	0.05

The grain-size exponent q corresponds to the configuration field `aexp`. The configuration excludes $R_{\mathrm{out}} < R_c$, while parameter-file generation requires $R_{\mathrm{in}} < R_{\mathrm{mid}} < R_{\mathrm{out}}$. Zone masses, shared dust properties and the taper exponent are derived quantities and do not increase the dimensionality.

Foreground reddening is treated separately as a nuisance parameter. For each trial containing an SED comparison, the code minimises the photometric discrepancy over

$$0 \leq E(B-V) \leq 5 \text{ mag}, \tag{5}$$

using the supplied Cardelli-law table. This one-dimensional fit is performed inside the objective evaluation and is additional to the 18 SMAC parameters. In the experiment without SED constraints, the current scorer leaves $E(B-V) = 0$.

3.2 Observational experiments

Each experiment uses one fixed combination of observations (Table 5).

Table 5: Observational constraints included in each experiment. PDI denotes all three polarimetric bands: V, I and H.

Experiment directory	SED	Interferometry	PDI	ALMA
`sed`	✓	–	–	–
`sed_interf`	✓	✓	–	–
`sed_pdi_unres_corr`	✓	–	✓	–
`sed_alma`	✓	–	–	✓
`sed_int_pdi`	✓	✓	✓	–
`sed_int_alma`	✓	✓	–	✓
`sed_pdi_alma`	✓	–	✓	✓
`int_pdi_alma`	–	✓	✓	✓
`sed_all_pdi_unres_corr`	✓	✓	✓	✓

For interferometry, the model uses one image per band at 1.65, 2.2, 3.5 and 10 μm. The fitted quantities are squared visibilities for PIONIER, GRAVITY and MATISSE L, and *correlated flux for MATISSE N*, as selected by the current data loader. Closure phases are diagnostic outputs and do not enter the objective.

Polarimetric images are calculated at 0.55, 0.82 and 1.63 μm. All experiments that include polarimetry enable the unresolved-polarisation correction, using an aperture radius of three instrument pixels. The optional additional overresolved interferometric component is disabled.

3.3 Objective function

SMAC minimises the sum of the active observable terms:

$$J = J_{\mathrm{SED}} + \sum_{b} J_{\mathrm{interf},b} + \sum_{k} J_{\mathrm{PDI},k} + J_{\mathrm{ALMA}}. \tag{6}$$

Only terms corresponding to the observations selected for an experiment are included.

Table 6: Contributions to the optimisation objective.

Observable	Contribution
SED	Uncertainty-weighted squared residuals, divided by $N_{\mathrm{SED}} - 1$, after fitting reddening.
Interferometry	Sum of instrument-specific uncertainty-weighted squared residuals, each divided by $N_b - 1$.
PDI	Weighted discrepancies in polarised-light fraction, normalised radial profile and six differential-sector measurements.
ALMA	$100 \sum (I_{\mathrm{obs}} - I_{\mathrm{mod}})^2 / \sum I_{\mathrm{obs}}^2$, within the comparison aperture.

For each polarimetric band,

$$J_{\mathrm{PDI}} = 10^5 (\Delta f)^2 + 10^4 \ \mathrm{mean}_j[(\Delta p_j)^2] + 5 \times 10^3 \sum_{\ell=1}^{6} (\Delta d_\ell)^2. \tag{7}$$

Here, f is the positive-Q_ϕ flux divided by total intensity, p_j is the normalised signed-Q_ϕ radial profile, and d_ℓ denotes the differential-sector quantities. These empirical terms make the combined objective a composite discrepancy score; it does not have the interpretation of a single reduced chi-square or a calibrated posterior likelihood.

3.4 Search algorithm and reproducibility

The search uses SMAC3 with a random-forest surrogate and a scrambled Sobol initial design. Its configured controls are listed in Table 7.

Table 7: Current optimisation controls for the nine seed1 experiments.

Setting	Value
Trial limit per experiment	1000
Parallel workers	5
Whole-trial walltime limit	1500 s for `sed`; 3600 s for the other eight experiments
Timeout cost	`none`; no fit score assigned to timed-out trials
Warm start	`sed` imports its previous run history; none configured for the other eight experiments
Requested Sobol initial configurations	256
Initial-design seed	101
Optimiser seed	Randomly generated; no explicit `-seed` supplied
Objective treatment	Stochastic: `deterministic=False`
Fidelity stages per experiment	One

Setting	Value
Minimum and maximum budget	Both 1
Photon-count changes during optimisation	None

Although the code uses SMAC's `MultiFidelityFacade`, these particular experiments have a single stage. Every evaluation within an experiment therefore uses the same selected observations and numerical settings. The directory name `seed1` identifies the experiment set; it does not imply an optimiser seed of 1. The initial-design seed also does not set an explicit MCFOST random seed.

The current `sed` launch script uses `-warmstart` with `sed/slowed_by_time/optimization/47597/runhistory.json`, relative to the seed1 directory. The walltime limits cover the entire trial, including MCFOST and scoring.

A Source provenance and verification

All paths below are relative to `/fred/oz061/kandrych`. The representative run directory is `smac/1_tapered_2zones_dubrulle/lower_precision/seed1/sed`. The physical YAML configurations in all nine experiment directories were verified to be semantically identical, and their MCFOST templates were verified to have identical content.

Table 8: Source files supporting this description.

Topic	Source
Free parameters and constants	`configuration_tapered.yaml` in each experiment directory
Fixed stellar, physical and numerical settings	`simulation.para` in each experiment directory
Launch options and initial-design seed	`run_smac.q` in each experiment directory
Selected observations	`fidelity.txt` in each experiment directory
Parameter generation, continuity and ties	`modelling_optimisation/lib/obriy_mcfost.py`: `write_mcfost_paramfile`, `_2zone_cont_calc_dmass`, `_apply_selected_second_component_fractions` and `ParaFile.save`
SED secondary and foreground reddening	`modelling_optimisation/lib/obriy_sed.py`: `add_blackbody_component`, `fit_sed_reddening` and `chi2_SED_with_reddening`
Interferometric secondary and score	`modelling_optimisation/lib/obriy_interferometry.py`: `calc_observables_with_secondary`, `monochromatic_chi` and `oi_container_chi2`
Interferometric data selection	`modelling_optimisation/lib/obriy_interferometry_data.py`
Polarimetric score	`modelling_optimisation/lib/obriy_polarimetry.py`: `measure_pdi_constraints`, `pdi_constraint_loss` and `compare_pdi_constraints`
ALMA score	`modelling_optimisation/lib/obriy_alma.py`: `compare_alma_images`
Combined objective	`modelling_optimisation/lib/obriy_mcfost.py`: `load_and_score_outputs`

Topic	Source
Optimisation and initial design	`modelling_optimisation/SMAC3.py`: `main`
Fidelity-stage interpretation	`modelling_optimisation/lib/obriy_fidelity.py`

The derived quantities were checked by generating a temporary parameter file with the current writer and the default configuration. This verified the contiguous boundaries, mass split, shared scale heights and flaring, outer taper tie, distance override, and shared silicate–iron mixture. No modelling code or run configuration was changed, and no radiative-transfer or optimisation run was launched during this inspection.